Team 10 - Detecting Narrative Reframing on X in the 2024 U.S. Election

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1. Introduction

According to a Pew Research study in 2026, 59% of Americans report that they get their news from social media [1]. While social media has allowed people to communicate and organize grassroots movements such as #MeToo and #OccupyWallStreet, it has also facilitated the spread of misinformation. On platforms like X, people share news and argue about what it means. We call the angle used to discuss an issue a ‘frame’. For example, immigration can be framed as a jobs issue one week and as a crime issue the next. When the dominant frame for an issue changes, we call it a narrative shift. During the US presidential elections, many users shaped online narratives around various topics such as foreign policy and immigration and shifted the narratives both intentionally and unintentionally. Past studies of the 2016, 2020, and 2024 elections examined how important events changed online conversations. These studies start from a known event, such as a debate, and measure how the conversation reacted. Our project tries to detect shifts in an issue's dominant framing using tweet data across every week of the 2024 U.S. election campaign. This allows us to find shifts whether or not a major event occurred and then predict them from features of the surrounding discourse.

A detector like this could help researchers and journalists see when a political conversation is changing. Narrative shifts that can’t be explained by a news event are important because they may come from coordinated accounts trying to push a narrative or from slow-building stories. This project aims to build the detector that later studies could use to analyze the cause of these narrative shifts. There is limited research done on narrative shifts across an entire campaign, which is what our project sought to analyze. A detector that only looks near known events may be detecting only a portion of the narrative shifts, and we wanted a method that treats every week the same way. To study this, we tracked how each issue was framed from week to week using two topic models, Latent Dirichlet Allocation (LDA) and Non-negative Matrix Factorization (NMF). We then used weekly changes in those topics as input features for a classifier that predicted when a narrative shift occurred.

Prior works either detect topic change without training a classifier or train classifiers on tweet volume. Our contribution is to track framing issue by issue and week by week across a whole campaign on tweets, with a shift score that separates new wording from shifting attention, and to use those scores to predict labeled narrative shifts.

For the unsupervised portion, we used our LDA and NMF models to check whether our five issues (immigration, economy, abortion, democracy, and foreign policy) emerged as topics, tagged tweets by issue using keyword patterns, and then fit a separate NMF model for each issue in each week. We then matched each week’s frames to the previous week’s and computed a shift score that combines how much the wording of matched frames changed with how much attention moved between frames. These scores then became features for the supervised model in Part A.

From this, we found that with LDA and NMF, we saw our 5 issues only once candidate names were removed and once we raised the topic count. We also found that only about 13% of tweets mention any of the five issues. Weekly narrative shift scores were highest for abortion and democracy; these two issues stayed on top across all six sensitivity settings (though they were also our smallest issues). We also found that shifts came mostly from new wording within existing frames, not from attention moving between frames. Additionally, the peaks lined up with events, for example, the June debate and Biden dropping out of the presidential race. Some shifts matched campaign events, and others, such as abortion in October, did not match any obvious event.

The supervised learning portion of the project predicted whether each issue in each week of the campaign (an issue-week) contained a narrative shift. Because the tweet corpus has no narrative-shift labels, we created training labels with Claude Sonnet 5.5. The model read a sample of tweets for every issue-week and labeled it as shift 1 or no shift 0, using a majority vote over three runs. We then compared a blind hand check of a random sample of rows against those labels. We then trained three model families, Logistic Regression, Gradient-Boosted Trees (GBT), and a Support Vector Machine (SVM), to predict those labels from the Part B change scores and tweet volume. Lastly,

we compared each model against two baselines using 5 time-based folds to determine whether any model family could separate shift weeks from other weeks.

Supervised learning found that GBTs had the highest mean PR-AUC (0.38 0.16) and were above both baselines, but none of the models were significantly better than the chance level. We saw an increase in mean PR-AUC from 0.32 to 0.41 after we added the Part B unsupervised features, but most of that improvement came from a single test fold that contained only one shift. Small changes in a few predictions can change the fold-level scores greatly with our 137 issue-weeks and 22 labeled shifts (we dropped 13 rows that were missing a shift score or lag feature). Due to this, we treat these results as uncertain rather than as evidence that the framing features predict shifts.

2. Related work

Cinus et al. previously conducted a study, Exposing Cross-Platform Coordinated Inauthentic Activity in the Run-Up to the 2024 U.S. Election [2]. While this study also examined the 2024 election, our project differs in several ways. The study was concerned with identifying networks of coordinated accounts across X, Facebook, and Telegram from May to June 2024. However, our project used weekly topic models to find how the framing of issues changed over time from May to November, and then trained a supervised classifier to predict which issue-weeks contained a shift.

Another study was conducted by Lange et al. in "Narrative Shift Detection: A Hybrid Approach of Dynamic Topic Models and Large Language Models" [3]. In this study, they analyzed Wall Street Journal articles from 2009 to 2023, performed topic modeling using RollingLDA, a version of LDA that tracks topics over time, and employed an LLM to explain topical changes and determine if a narrative shift had occurred. Our project differs in that while an LLM was used to label narrative shifts, we went further by using supervised learning methods that try to predict those labels from topic-change scores and tweet volume.

Farr et al. analyzed the discourse on X around the 2026 U.S. capture of Nicolás Maduro [4]. In their study, they also looked at narrative drift by measuring how far each topic cluster moved in meaning over time. However, their study focused more on analyzing how the structure of the topic clusters evolved after a large event and how the clusters became less noisy after discourse around narratives formed. Our project looked at the evolution of topics across every week of the campaign, whether or not a major event occurred, and added shift labels and a supervised classifier.

This project is new work for this course. It does not extend a Milestone I project or any earlier course projects.

3. Data Source

Our data comes from the USC X 24 US Election Twitter/X Dataset available on GitHub [3]. This dataset contained ~46 million tweets subdivided into 50,000 compressed CSV (.csv.gz) chunk files that were posted on X between May 1, 2024, and November 30, 2024. After keeping only original (non-retweet) English tweets, a representative sample of about 15,000 tweets per week across 31 weeks, resulting in a total of 465,146 tweets, was created for analysis. We ran these tweets through a preprocessing pipeline to clean the data and standardize the schemas. In order to accommodate ease of distribution amongst team members, the cleaned sample was stored as a Parquet file. The primary variable used in our analysis was the content of the tweet itself (rawContent). We also used the posting date (date) to assign tweets to weeks, the language code (lang) to keep English tweets, the retweet column (retweetTweet) to drop retweets, and the tweet link (url) to get each tweet's ID and remove duplicates.

4. Feature engineering

From the original data in the USC X dataset on GitHub, we applied several steps before using it in our analysis. First, we iterated through each file and performed several preprocessing steps, including standardizing column headers, removing duplicate tweets, removing retweeted tweets, and filtering for English-language tweets. During this step, we also processed the tweet text. We did not want to use "mentioned" users in topic modeling; however, if

the mentioned user appeared in a sentence, we kept them in the tweet. Additionally, the original web scraping inconsistently kept apostrophes in contracted words, which we corrected in this step. Finally, we normalized nonbreaking spaces and newline characters, and removed HTML-entity text such as “amp” using the stopword lists.

For the unsupervised models, we started from the cleaned tweet text and dropped tweets with fewer than 5 words, since they gave too little text to model. LDA used raw word counts, and NMF used TF-IDF (Term Frequency-Inverse Document Frequency) weights. We removed standard English stopwords and, for the 20- and 30-topic runs, candidate names, party labels, and filler words. We tagged each tweet by issue with keyword patterns (one true/false column per issue), then fit weekly NMF models on each issue’s tweets with 5 frames each. From those models, plus the 5-topic NMF model, we built a table with one row per issue-week: 155 issue-weeks, 149 of which had enough tweets to score. Its columns were:

- week, week start and end dates, issue
- tweets about the issue that week, total tweets that week, and days of data in the week
- issue share of the week’s tweets, and its change from the previous week
- content shift, proportion shift, number of new frames, and the shift score
- the share of tweets falling into each of five broad topics from the 5-topic NMF model (Biden, Trump, Harris, and the MAGA movement, plus one on ‘partisan talk’)

The supervised models use 17 features: 11 from the Part B issue-week table and 6 that we added to measure tweet volume, sampling noise, and each issue’s recent history. We added additional features because Part B change scores cannot tell the difference between a narrative shift and a week with more tweets, normal sampling variation, or the amount of normal weekly change for that issue. We organized these features into four groups that include 3 volume, 7 frame change, 5 topic mix, and 2 past week features (see Table A1 in the Appendix for the full list). Because each weekly sample was capped at about 15,000 tweets, we couldn’t distinguish between loud and quiet weeks. To measure volume, we estimated each issue’s raw data amount as orig_tweet_count = weekly_orig_total × issue_share, where issue_share is the issue’s share of the weekly sample. We then divided this count by the number of days with data to correct for gaps in collection and the 4-day final week, and applied a log transform to reduce right skew: log_orig_per_day = log(orig_tweet_count / days_with_data).

The noise floor gives us a baseline so we can avoid false positive narrative shifts. To estimate the sampling noise, we randomly split each issue-week’s tweets into two halves, scored the change between halves with the same method we used between weeks, and averaged over 20 random splits. We define shift_minus_noise as the shift score minus the noise floor. Issue-weeks with fewer than 100 qualifying tweets (so that each half meets the 50-tweet minimum) have no noise floor: we found this in 16 of the abortion weeks and tracked them with the column “noise_missing”. To represent each issue’s recent history, we added two lag features, lag1_shift and lag2_shift, which are the issue’s shift_score from one and two weeks earlier, and shift_zscore, which standardizes the current shift_score using the issue’s past weeks only. For shift_zscore, we required at least three past weeks and clipped the values to [-3, 3], because early weeks with very low variance produced extreme values such as -11.67.

Of the 150 issue-weeks we labeled (weeks 1-30; see Part A), we dropped 13 that were missing a shift score or lag feature, which left us with 137 rows and 22 shifts. Week 0 has no shift scores because there is no prior week to compare to it, so weeks 1-2 of every issue are missing at least one lag feature (10 rows). One abortion week had only 45 tweets, below our 50-tweet minimum, so we didn't score it, and abortion weeks 29-30 are missing the lag features that depend on it (3 rows). The remaining missing values in shift_minus_noise (12 rows) and shift_zscore (5 rows) are handled with a median imputation in the Logistic Regression and SVM pipelines, which are fit on each training fold only. The HistGradientBoostingClassifier handles missing values on its own.

Part B. Unsupervised Learning

We present our unsupervised learning first because its outputs are the inputs to our supervised learning.

B.1 Methods description

Our goal was to determine whether we could detect shifts in narratives during the 2024 election cycle. To do that, we first needed to identify which issues people were talking about, then track how they discussed each issue and how that changed week to week. The pipeline has four steps:

1. We fitted LDA and NMF topic models on the full 465k tweet sample to check whether our five issues (immigration, economy, abortion, democracy, and foreign policy) appear as topics.
2. We tagged our sample tweets by issue using keyword patterns.
3. For each issue and each week, we fit a small NMF model to find that week's "frames," groups of words that tend to appear together.
4. We matched frames across consecutive weeks, then computed a narrative shift score. These weekly shift scores became the features passed to the supervised model as a table with one row per issue-week (155 rows: 31 weeks × 5 issues, 149 scored).

LDA and NMF are both topic modeling algorithms; however, they work very differently, so we compared both. LDA is probabilistic; it treats each tweet as a mixture of topics and each topic as a distribution over words. NMF is a matrix factorization method; it splits a TF-IDF matrix into nonnegative topic and tweet weights, with no probability model. We expected NMF to produce cleaner and more coherent topics on short tweets because TF-IDF would have down-weighted common words. We expected LDA to produce more distinct topics with less overlap. We tested both expectations in Part B.2. We used LDA for the discovery step and NMF to track weekly frames, since it is more deterministic with a fixed seed and refits quickly about 150 times. We can compare its topic vectors directly using cosine similarity.

LDA used raw word counts (CountVectorizer), since the model assumes counts, and NMF used TF-IDF weights. At first, we ran 5 topics with the default stopwords. The topics mainly focused on the candidates, led by 'harris, kamala, president, vp, vote' and 'biden, joe, president, administration, hunter,' while issue words like 'border' or 'Ukraine' barely appeared. For the 20- and 30-topic runs, we removed candidate names, party labels, and filler words, and lowered max_df from 0.95 to 0.5. This drops words that appear in more than half of tweets, since common words show up in every topic and don't help distinguish topics.

With 5 topics, both models grouped tweets by candidate (Trump, Harris, Biden) rather than by issue. Only about 13% of tweets mention any of our five issues, so candidate talk dominated a great deal of the results. LDA with 20 topics found all 5 issues, but we labeled the topics by hand, and the abortion and democracy topics were mixed or weak. To reliably track tweets over time, we used regex patterns for each issue. We then fixed false matches, such as the word "nato" being found within "senator" and "roe" being found within "heroes," using word boundaries, and measured precision by hand on 100 tweets (Part B.2). We dropped healthcare because it never appeared as a topic.

For the discovery step, we compared LDA and NMF at 5, 20, and 30 topics, using '*top words'* charts and coherence to judge them. For the weekly step, we varied the number of frames per issue (K = 3, 5, 7), the frame-matching threshold (0.1, 0.2, 0.3), and the period length (1 week v. 2 weeks). We set K = 5, threshold = 0.2, and weekly periods as defaults because they gave enough frames to show change while keeping enough tweets per frame. We chose weekly analysis because it gives a clearer sense of when and how narratives shifted. We required at least 50 tweets to fit our issue weeks; we skipped issue weeks with fewer tweets. We fit LDA with scikit-learn's default online variational learning (up to 10 passes) and NMF with NNDSVD initialization and up to 500 iterations. Both used random_state = 42, allowing the results to be reproduced.

For each issue, we fit a separate NMF model on each week, then compare each week's frames to the previous week's using cosine similarity. We matched frames one-to-one with the Hungarian algorithm (linear_sum_assignment). We count a pair as the same frame only if its similarity is at least 0.2. Frames that don't pair up above that cutoff are counted as new frames. We then computed two measures. Content shift is 1 minus the average similarity of the matched frames, so it is high when the wording inside a frame changed. Proportion shift is the Jensen-Shannon distance between the two weeks' frame shares, so it is high when attention moved from one frame to another. The shift score is the average of the two, from 0 (no change) to 1 (complete change). The number of new frames is

reported separately. We used cosine similarity because it ignores tweet volume, and Jensen-Shannon because it is symmetric and bounded.

Originally, we considered BERTopic, a clustering that was embedding-based; however, due to the size of the data, this would be a very time-consuming and much more extensive approach, though it would've given more intuitive interpretations for the tweets that perhaps didn't explicitly mention a word but still discussed a topic we would've been interested in. We chose LDA and NMF because they are interpretable (every topic is a word list we can read and check), cheap to refit for every issue-week, and standard baselines. We list embeddings as future work.

B.2 Unsupervised Evaluation

In this section, topics have no labels, so we could not score them against any right answer. Instead, we used four checks, each measuring something different: Topic coherence (NPMI, normalized pointwise mutual information) measures whether a topic's top 10 words appear in the same tweets. A topic like 'border, illegals, immigrants' should score higher than a mix of unrelated words, giving us a better indicator of relatability between words. Topic diversity is the share of unique words across all topics, and low diversity means topics repeat each other. For example, with 5 topics, several topics were led by the same names, like 'trump,' 'biden,' and 'president.' Issues covered counts how many of our five issues appear in a topic's top words, indicating whether the model finds the issues we care about. Keyword precision is the share of tweets tagged with keywords that actually pertain to the issue. We manually checked 100 tweets (20 per issue); for example, a tweet about the 'Russia hoax' contains a foreign policy keyword but isn't about foreign policy. Coherence and diversity measure topic quality. Issues covered and keyword precision measure whether the output is useful for our question. In Table 1, we report all four because no single one captured the insights we wanted to understand. It was important to identify the major themes in these tweets to understand what people found important to discuss.

Model	Topics	Coherence	Diversity	Issues covered (of 5)
LDA	5	0.035	0.66	2
NMF	5	0.076	0.78	0
LDA	20	0.090	0.94	5
NMF	20	0.107	0.79	4
LDA	30	0.079	0.92	4
NMF	30	0.110	0.75	5

***Table 1**. LDA v NMF (465k Tweets, topic rows used default stopwords and the 20/30 topic rows used the name list)*

The best run from each family was LDA at 20 topics and NMF at 30. NMF was more coherent at every topic count, and LDA was more diverse. We used both for different purposes: LDA at 20 topics gave us the most distinct topics and captured all five issues, so we used it to confirm the issues appear in the data. NMF gave us tighter, more related word groups, and it's quick and repeatable to refit for every issue and week, so we used it to track how each issue changed over time. At 5 topics, we grouped both by candidate and found few or no issues. We then removed candidate names, party labels, and filler words for the larger runs.

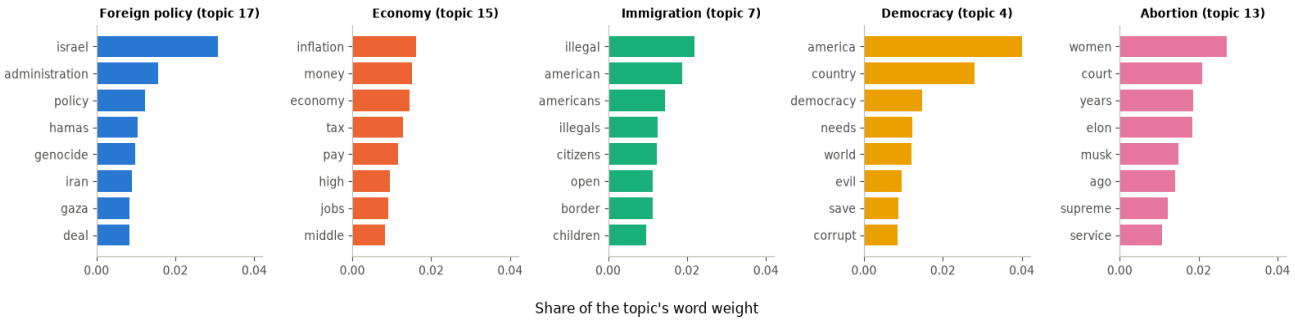


Figure 1

Figure 1 above shows the top words of the five LDA topics (20-topic model) that we read as issues. Bars show each word's share of the topic's word weight. shows the top words of the five LDA topics we read as issues. Immigration, economy, and foreign policy are clear. The abortion topic is mixed with court and Musk words, and the democracy topic is weak (America, country, democracy). Figure 2 shows how many tweets each topic dominates. Each issue topic is the strongest topic for only 3.7-6.8% of tweets; the other topics are candidates, parties, debates, and reactions. LDA forces every tweet into one topic, so these shares can't be compared to the 13% keyword rate.

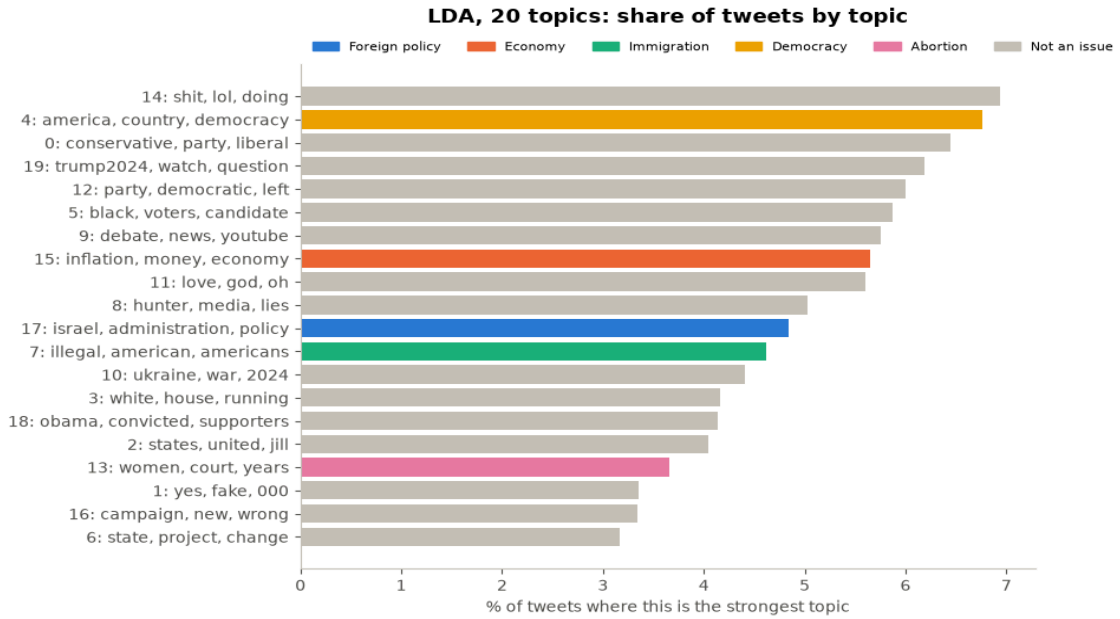


Figure 2*. Share of tweets for which each LDA topic is the strongest topic (20 topics, 465k tweets). Colors mark topics we read as issues; gray topics are not issues.*

Abortion has the highest average shift score (0.387), followed by democracy (0.318), foreign policy (0.302), immigration (0.277), and economy (0.274). Peaks line up with events: foreign policy and economy around the June 27 debate; economy, foreign policy, and immigration after Biden dropped out July 21; immigration in late September; and democracy and abortion after the election. Content shift (0.40-0.63) was roughly three to four times the proportion shift (about 0.14). As mentioned previously, the change seems to come mostly from new wording within the same frames, not from attention shifting between them. Attention and shift were only weakly related (Spearman: -0.37 to 0.16). Figure 3 shows the weekly shift score for each issue as a heatmap, and Figure 4 shows the same scores as lines, with each issue's three highest weeks marked. Abortion and democracy are our smallest issues, so part of their higher scores may reflect noise.

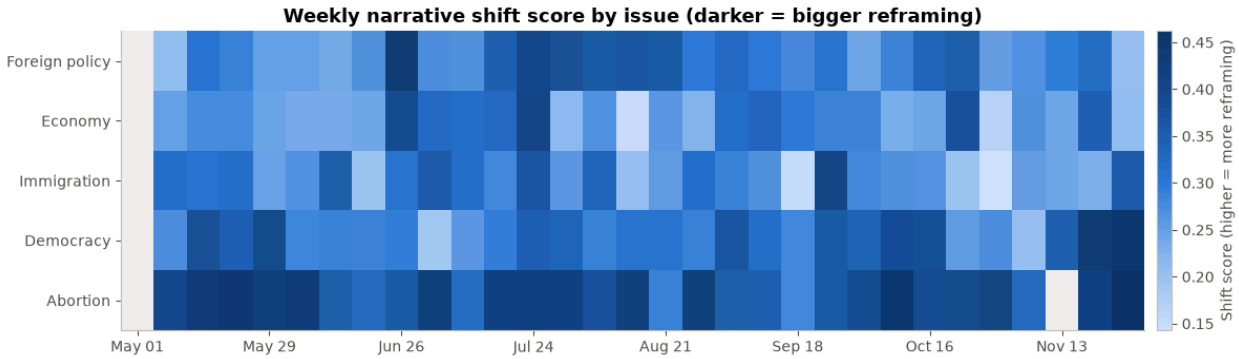


Figure 3. *Weekly narrative shift score by issue (darker = more reframing). Light gray cells are weeks with no score.*

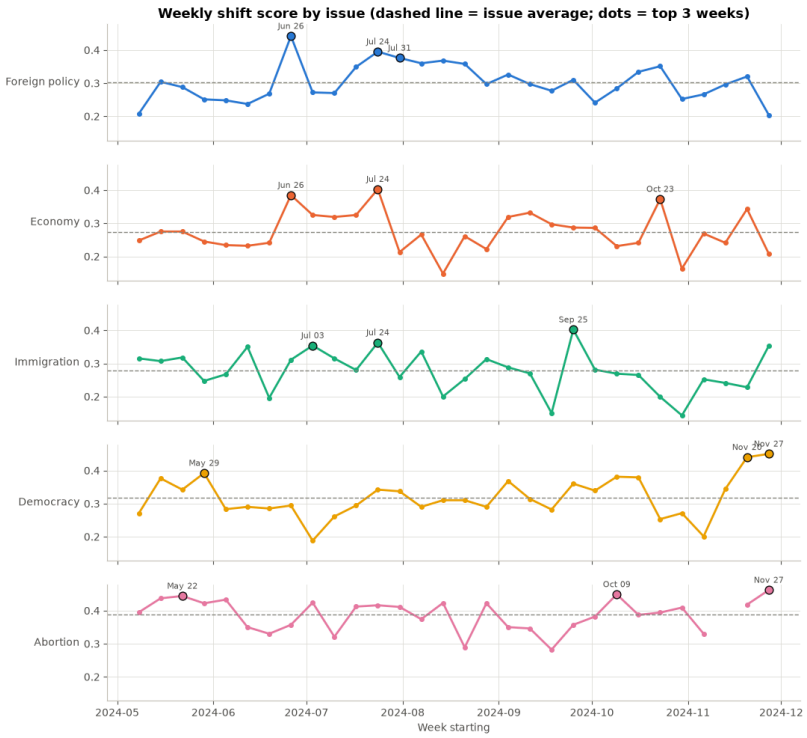


Figure 4. *Weekly shift score by issue. The dashed line is the issue's average, and the dots mark its top three weeks. The gap in the abortion line is a week with too few tweets to score.*

B.3 Sensitivity Analysis

We reran the weekly pipeline across six settings: K = 3, 5, 7; threshold = 0.1, 0.2, 0.3; and 1-week vs. 2-week periods. Abortion and democracy were the top two in all six settings, and the overall issue ranking stayed close to baseline (Spearman 0.9-1.0). The weekly pattern held up when we changed the threshold (median Spearman 0.83-0.91) but changed more when we changed K (0.42-0.44). So we trusted issue rankings and major peaks, and we didn't read much into small weekly changes.

Setting	Foreign policy	Economy	Immigration	Democracy	Abortion	Ranking v. Baseline	Weekly pattern v. Baseline
Baseline (K=5, threshold 0.2, 1 week)	0.302	0.274	0.277	0.318	0.387	1.0	1.0
K=3	0.251	0.251	0.244	0.305	0.340	0.9	0.42
K=7	0.290	0.279	0.295	0.327	0.376	0.9	0.44
Threshold = 0.1	0.321	0.294	0.300	0.354	0.417	1.0	0.83
Threshold = 0.3	0.278	0.263	0.269	0.300	0.365	1.0	0.91
2-week periods	0.247	0.246	0.254	0.279	0.332	0.9	n/a (different periods)

***Table 2**. Average shift score by issue under the six settings and how closely each setting matches the baseline.*

To check how well the keywords worked, we pulled a random sample of 100 keywords tagged in the tweets (20 per issue) and marked each one as 1 if it was really about the issues and 0 if it was not. The overall precision score was 0.85. Results showed scoring was highest for immigration and abortion, given a score of 0.95, for democracy 0.85, economy 0.80, and foreign policy 0.70. Foreign policy was lowest because country names were associated with smears and passing mentions such as: 'Russia Hoax.' Prior to this check, we had fixed nato matching inside 'senator' and roe inside 'heroes,' which cut foreign policy tweets by about 9% and abortion by about 8%. With only 20 tweets per issue, a few borderline calls could shift any issue's score noticeably.

Part A. Supervised Learning

A.1 Methods description

Workflow: (1) Label each issue-week. (2) Join the labels with the features from Part B and the features from the volume and noise floor. (3) Set two baseline comparison points that any model could outperform. (4) Train three model families with week-based cross-validation. (5) Tune settings with nested cross-validation. (6) Study the best model using ablation, feature importance, sensitivity, tradeoff, and failure analysis.

Our X dataset didn't come with labels for narrative shift, so we created the labels using an LLM. We wrote a script that built a packet for each of the 5 issues in weeks 1-30 (150 issue-weeks). Each packet has up to 30 random tweets about the issue from last week and up to 30 from this week. The packets purposely leave out the dates, week numbers, and scores so the labels can't copy our features. We then wrote a prompt that Claude Sonnet 5.5 followed to read the packets and label each issue-week as shift (1), no shift (0), or unsure. The prompt discusses the rules for the LLM to follow, provides examples we labeled by hand, and prompts it to only judge from the tweets in the packets so that it doesn't include external date information to decide if a week was likely a shift or not based on key political events. The rules included labeling an issue-week as a shift if 1 in 4 tweets (or about 8 of the 30 tweets in a packet) frame the issue in a new way, have a new target to blame, or have a new story. The prompt also states what does not constitute a shift, such as having more or angrier tweets or a new event that's discussed the same way as previous events before.

We had the LLM label the packets three times because during trial runs its decision would change, so we decided to do multiple runs and take the majority vote (2 out of 3). This resulted in the LLM agreeing 3 out of 3 on 140 out of 147 rows with no row labeled as "unsure." To verify the LLM labels, we considered having multiple teammates

hand-label 20-30 packets each and use Cohen's kappa to compute an agreement score with the LLM, which was done in similar research work, but due to time limitations of the course, we decided to only have the supervised lead label 10 random packets, 5 shifts, and 5 no-shifts in random order with the model's label hidden. We agreed with the model 9 out of 10 times. The disagreed packet (abortion, week 8) was an argument that was framed the same way but by a new person, and we considered that as no shift. We used our label as the final judgement. The labels result in a 16% shift rate (24 shifts out of 150 rows).

We use three model families, each of which learns differently. Each row in our data is one issue-week. We use 137 rows with 22 shifts because we dropped 13 of the 150 labeled rows that were missing a shift score or a lag feature. This is the first 2 weeks of data (10 rows) and abortion weeks 28-30 (3 rows) because week 28 didn't have enough sampled tweets to score, and weeks 29 and 30 use that missing score as a lag. All three models use class_weight= "balanced" so that the shift rows count more during training and the model doesn't do well by always labeling as no-shift. These models were chosen because we needed simple models that would be less likely to overfit on our small dataset. Initially, we considered doing LSTM and Transformer models, but those would need thousands of examples. The Logistic Regression and SVM models standardize the features because both are sensitive to feature scaling.

1. Logistic Regression (probabilistic, linear): We chose this model as our starting point. It has few parameters to learn, so it's less likely to overfit our 137 rows. This model multiplies each feature by a weight, adds up the weighted features, and turns the total into a probability of a shift. The results are also easy to explain because the weights show which features influence shift weeks.
2. Gradient-Boosted Trees (tree-based ensemble, non-linear): This model was chosen because it can find non-linear patterns and interactions between features. This is helpful for issue-weeks with few tweets because those rows can get high shift scores by chance, whereas a high shift score may only be a real shift when the issue has enough tweets. We limit the trees to 2-3 levels to prevent overfitting. This model builds many small decision trees sequentially, and each new tree learns from the errors of the trees before it. We used scikit-learn's HistGradientBoostingClassifier because it can handle missing values, which two of our features have. The other two models need the missing values filled in.
3. Support Vector Machine with an RBF kernel (non-probabilistic, margin-based): We chose this because it learns from the distance between rows, instead of adding up weights or splitting on rules like the other two models. It finds the boundary with the widest margin from the closest shift and no-shift rows, and the radial basis function (RBF kernel) lets that boundary curve instead of being straight. The boundary depends mostly on the rows closest to it, which are the hardest to tell apart (the support vectors). This can help it to work well when there are few rows.

For our baselines, baseline 1 always predicts no shift, while baseline 2 tests whether a trained model performs better than the Part B score and uses the unsupervised shift score with no learning.

We turned scores into binary predictions so we could report precision, recall, and F1. First, we tried a decision threshold that we tuned on the training weeks, but it didn't transfer to the test weeks: the test scores fell in a different range since there are not very many training rows, and the threshold labeled either all or none of the test rows as a shift. We then used a top-1 prediction rule, where the issue with the highest score is predicted as a shift for each week. This rule has challenges because in reality there can be more than one shift in a week. We discuss this in the failure analysis, and we test top-2 and top-3 in the sensitivity analysis.

We used a 5-fold time-based cross-validation with scikit-learn's TimeSeriesSplit. We split by week so all five issues from a week are always in the same fold. Each fold trains on earlier weeks and tests on later weeks, with a one-week gap between them because neighboring weeks tend to be similar, and this separation reduces the dependence between the training and test data. Since this is temporal data, we made sure not to use random splits because it could introduce leakage by training on later weeks to predict earlier ones. We then dropped rows that didn't have enough historical data (our lags) and were left with 137 rows from weeks 3-30 with 22 shifts in total. This left each test fold with 17-20 rows and 1-7 shifts.

We used nested cross-validation for hyperparameter tuning. First, we used a grid search over three time-based inner folds within each outer training set. Then we selected the combination with the highest mean PR-AUC and retrained

it on the full outer training set. Then we evaluated it once on the outer test fold to make sure the test weeks were not in the model selection. We kept the grids small (Table 3) since we didn't have many rows, and searching more combinations could lead to overfitting the inner folds.

Model	Settings tried	Combinations
Logistic Regression	C (regularization strength) = 0.01, 0.1, 1, 10	4
Gradient-Boosted Trees	Max_depth = 2, 3; learning_rate = 0.5, 0.1; min_samples_leaf = 3, 5, 10	12

Table 3. *Hyperparameter Grids Table.*

A.2 Supervised Evaluation

A.2.1 Overall Results

Our main evaluation metric is PR-AUC and is computed as average precision with scikit-learn's average_precision_score. We use this instead of ROC-AUC because it's more informative when the positive class is rare. The shift rate in the test folds is about 0.18, which equals a random classifier's expected PR-AUC. We also report precision, recall, and F1. We didn't use accuracy because of the class imbalance. Each of our test folds only has about four predicted shifts, which results in F1 having higher variance across folds than PR-AUC because one additional true positive changes F1 greatly.

Model	PR-AUC (mean ± std)	F1 (mean ± std)	Precision	Recall	Pooled PR-AUC	Permutation p-value
Baseline 1: always no shift	0.18 ± 0.12	0.00 ± 0.00	0.00	0.00	0.18	-
Baseline 2: shift_score only	0.27 ± 0.15	0.20 ± 0.19	0.20	0.31	0.20	0.47
Logistic Regression	0.31 ± 0.22	0.14 ± 0.14	0.15	0.15	**0.38**	0.28
Gradient-Boosted Trees	**0.38 ± 0.16**	**0.28 ± 0.22**	**0.25**	**0.41**	**0.26**	**0.17**
SVM (RBF)	0.35 ± 0.17	0.17 ± 0.17	0.15	0.28	0.24	0.12
Logistic Regression (tuned)	0.30 ± 0.22	0.14 ± 0.14	0.15	0.15	-	-
Gradient-Boosted Trees (tuned)	0.35 ± 0.12	0.20 ± 0.23	0.20	0.21	-	-
SVM (RBF) (tuned)	0.31 ± 0.19	0.22 ± 0.15	0.20	0.35	-	-

Table 4. *Evaluation Metrics Table. Results across the 5 time-based folds (mean ± standard deviation). Pooled PR-AUC combines all test rows before scoring. The p-value comes from 100 label shuffles.*

All 3 models perform better than the baselines on mean PR-AUC: Gradient-Boosted Trees (GBT) has the highest score (0.38), followed by SVM (0.35) and Logistic Regression (0.31). However, the ranking of the models is not stable because it depends on how PR-AUC is aggregated. The mean across folds ranks GBT first, but when all test rows are pooled into one group, Logistic Regression ranks first (0.38), and GBT decreases to 0.26. No model is significantly better than chance. To build a null distribution in a permutation test, we shuffled the labels 100 times and re-ran the full cross-validation each time. The shuffled labels averaged 0.27–0.30 for PR-AUC, the shift_score baseline scored at the chance level ($p = 0.47$), and about 1 in 6 shuffles scored as high as GBT ($p = 0.17$).

Hyperparameter tuning also did not help, since mean PR-AUC decreased by less than one standard deviation across the folds. It's also possible the inner fold scores could not tell the settings apart, since the selected hyperparameter changed from fold to fold. For example, Logistic Regression's C moved between 0.01, 0.1, and 10. In addition, no model found a shift in Fold 2 (weeks 15–18), likely because the features for weeks 16 and 17 describe a partial week; those weeks only had two or three days of data. We chose GBT as the main model for further analysis because it had the highest mean PR-AUC and tuning didn't improve it. We used its initial hyperparameter settings of

max_depth = 2, learning_rate = 0.1, and min_samples_leaf = 5. We chose GBT as the main model for further analysis because it had the highest mean PR-AUC and tuning didn't improve it. We used its initial hyperparameter settings of max_depth = 2, learning_rate = 0.1, and min_samples_leaf = 5.

A.2.2 Feature Importance and Ablation Analysis

Ablation: We trained GBT three times on the same folds with three cumulative feature groups.

Feature group	Features	PR-AUC (mean ± std)	F1 (mean ± std)	Fold PR-AUC (1 / 2 / 3 / 4 / 5)
A: volume only	3	0.32 ± 0.19	0.21 ± 0.13	0.32 / 0.20 / 0.47 / 0.06 / 0.53
B: A + topic features	15	**0.41 ± 0.13**	**0.35 ± 0.15**	0.45 / 0.23 / 0.31 / 0.50 / 0.55
C: B + past weeks (all)	17	0.38 ± 0.16	0.28 ± 0.22	0.47 / 0.13 / 0.32 / 0.50 / 0.48

Table 5. *Ablation study for Gradient-Boosted Trees.*

PR-AUC increased in 4 of 5 folds after adding the Unsupervised features (A to B), and PR-AUC increased from 0.32 to 0.41 and F1 from 0.21 to 0.35. Most of this comes from fold 4, which only has one shift. This one positive means an average precision of 0.50, which is rank 2 of 20, and 0.06 means about 17th. If we exclude fold 4, groups A and B have a similar PR-AUC of 0.38 and 0.39. Adding lag features didn't improve the score. PR-AUC increased in only 2 of 5 folds by 0.01-0.02. We kept group C as the main model because switching to group B after seeing the test-fold results would introduce optimism bias from using the test data for model selection.

Feature Importance: We used group permutation importance since we have multicollinearity in our features. We went with 30 repeats per fold (n_repeats = 30) by randomizing each group of related features together.

Feature group	Mean drop in PR-AUC	Std across folds	Folds with a drop	Feature group
Frame-change scores	0.057	0.078	4 of 5	Frame-change scores
Volume	0.024	0.053	4 of 5	Volume
Past weeks (lags)	-0.004	0.085	2 of 5	Past weeks (lags)

Table 6. *Grouped permutation importance for Gradient-Boosted Trees. Note: A negative decrease means shuffling the group did not hurt the model.*

The frame-change scores were the most important feature group, but no group's importance was consistent across the folds. The frame-change scores had the largest mean importance with 0.057, followed by volume with 0.024. The lag and topic-mix groups had averages close to zero and positive values in some folds and negative in others. Every group had a standard deviation across folds that was larger than the mean. This suggests the frame-change scores with the ablation carry most of what the Unsupervised part adds. No model is significantly better than chance; these results only describe what the model uses and not confirmed patterns in the data (Figure 5).

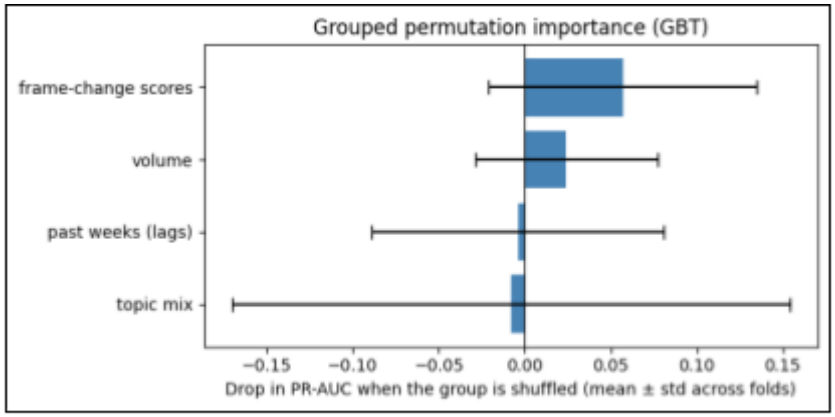


Figure 5. *Grouped permutation importance for Gradient-Boosted Trees: the decrease in PR-AUC when each feature group is shuffled (mean ± standard deviation across folds).*

A.2.3 Sensitivity Analysis

PR-AUC depended as much on which weeks were tested as on GBT's hyperparameters. We scored all 12 GBT combinations from the tuning grid on the same five folds, and overall, the scores increased and decreased together across the folds (Figure X). Mean PR-AUC had a range of 0.25 (0.30 to 0.55), which is larger than the main model's fold-to-fold standard deviation of 0.16. However, this again comes from the single shift in fold 4, which was only ranked 1st by the two highest-scoring combinations among that fold's test rows, and removing fold 4 decreases the range to 0.13 (0.30 to 0.43). The only hyperparameter with a consistent effect was min_samples_leaf: 5 scored best with or without fold 4, and 10 scored the worst. This is likely because the training folds had only 35–115 rows, so requiring 10 rows per leaf prevents the trees from isolating the small groups of shifts.

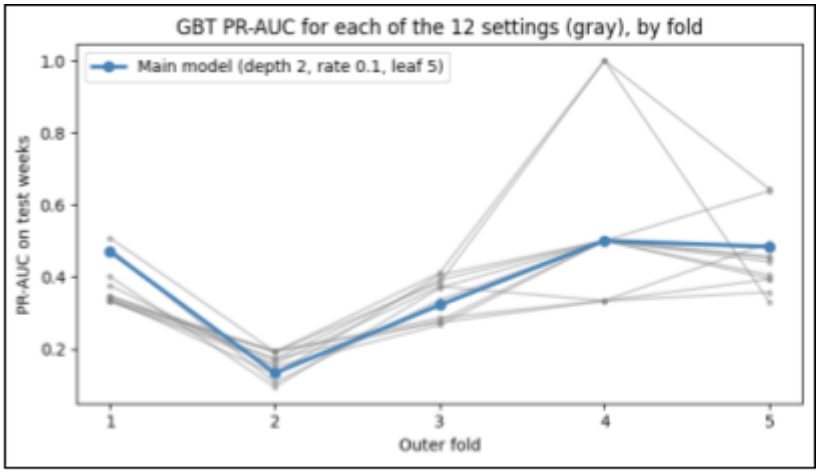


Figure 6. *Test–fold PR-AUC for each of the 12 GBT hyperparameter combinations (gray) and the main model (blue). The lines move together across folds.*

A.2.4. Tradeoffs

Precision vs. Recall: Predicting more shifts per week increases recall and lowers precision, and two predictions per week has a balance between the two (Table 7, Figure 7).

Shifts predicted per week	Predicted shifts	Shifts found (of 17)	Precision	Recall	F1
1 (main rule)	20	5	0.25	0.29	0.27
2	40	10	0.25	0.59	0.35
3	60	13	0.22	0.76	0.34
4	80	15	0.19	0.88	0.31
5 (every row)	97	17	0.18	1.00	0.30

Table 7. *Precision-recall tradeoff for Gradient-Boosted Trees on all 97 test rows pooled.*

Going from 1 to 4 predictions per week increases GBT's recall from 0.29 to 0.88 and reduces precision from 0.25 to 0.19. Predicting 2 per week doubles recall (0.29 to 0.59) with no loss in precision and gives the highest F1 (0.35). GBT performs better than Baseline 2 on both precision and recall at 1 to 3 per week. Predicting two per week is the better choice if missing a shift matters more than an extra false positive.

One of the reasons PR-AUC is the main metric is because it's the same under every option since it only depends on score ranking, and we kept top-1 as the main rule because we chose it before seeing these results. In comparison, F1 can predict every row as a shift and make F1 = 0.30, which is higher than GBT with the main rule (0.27), even though its precision equals the shift rate (0.18).

More training data improved PR-AUC in one fold but not in the other, so we can't measure this tradeoff with our data. We kept the test weeks fixed and trained on fewer or more of the most recent weeks. In fold 3, PR-AUC increased from 0.13 to 0.32 as the training window increased from 5 to 15 weeks, mostly when training shifts went from 3 to 12. In fold 5, there wasn't a clear pattern (0.33 to 0.53) and the smallest window scored highest. Each test

fold has only 3-4 shifts, and smaller windows are also more recent, so the effect of training size cannot be separated from these factors.

GBT had the best mean PR-AUC, but Logistic Regression is easier to explain and performs about as well. Logistic Regression had the best pooled PR-AUC (0.38 vs. 0.26 for GBT), and its coefficients show how each feature increases or decreases the predicted probability of a shift. The difference in mean PR-AUC (0.38 vs. 0.31) is smaller than the fold-to-fold standard deviation (0.16 for GBT, 0.22 for Logistic Regression), so Logistic Regression would be a reasonable choice.

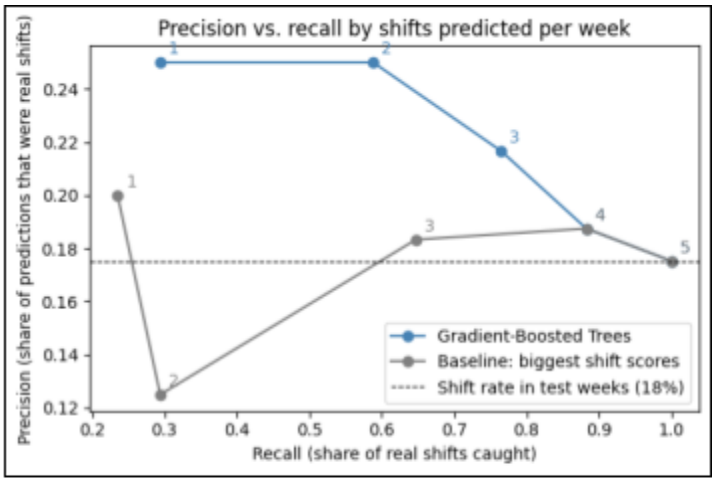


Figure 7. *Precision and Recall for GBT and Baseline 2 as the number of shifts predicted per week goes from 1 to 5. The dashed line is the test-fold shift rate (0.18).*

A.2.5 Failure analysis

Most errors come from the top-1 decision rule. The main model had 5 true positives, 65 true negatives, 15 false positives, and 12 false negatives across the 97 test rows. The top-1 rule accounts for 20 of the 27 errors (74%): 9 of the 12 false negatives are in weeks with two or more shifts, where the rule can find at most one, and 11 of the 15 false positives are in weeks with no shifts, where the rule still has to predict one (Table 8).

Condition	Rows	Shifts	Shifts found	Missed shifts	False alarms
Week has 2 or more shifts	20	12	3	9	1
Week has no shifts	52	0	0	0	11
Collection gap (3 or fewer days of data)	20	5	1	4	3
No noise floor (too few tweets)	8	1	0	1	2

Table 8. *Failure counts by condition (main model, 1 shift predicted per week).*

To explain individual failures, we used SHAP values [6], which split a prediction into additive contributions from each feature. We summed the SHAP values within the four feature groups used for permutation importance. Values are in log-odds.

Example	Result	Score	Shifts that week	Volume	Frame change	Topic mix	Lags
Economy, week of Nov 6 (week 27)	Missed shift	0.36 (2nd)	4	+0.52	+0.85	-1.02	+1.26
Abortion, week of Oct 30 (week 26)	False alarm	0.97 (1st)	0	-0.39	+3.02	+3.09	+0.27
Abortion, week of Aug 7 (week 14)	Missed shift	0.11 (2nd)	2	+0.07	+4.16	+0.81	-2.21

Table 9. *SHAP contributions by feature group for three failed rows.*

One of the failures we had was a decision-rule failure; we missed shifts in weeks where several issues shift at once. The top-1 rule predicts only one shift per week, so every other shift in that week counts as a false negative. For example, the economy issue in the week after Election Day was labeled a shift, but the model missed it. Tweets shifted from arguing over which candidate would manage the economy better to explaining why Harris lost on inflation and what Trump's economy would look like. The rule never predicted it because four of five issues shifted that week, and economy ranked second, and the topic-mix features score also decreased by −1.02 because the candidate-based topics did not describe economic framing well. This is a calibration problem, where the scores rank

issues within a week but don't mean the same thing across weeks, so a fixed threshold can't return zero, one, or several shifts per week. One way to fix this is to calibrate the scores. For example, we could add a week-level feature that measures change across all issues so that a threshold can set the number of predictions per week.

We also had feature failure, which comes from false positives caused by small-sample noise in the features. Abortion in the week before Election Day was labeled as no shift, but the model ranked it first with a score of 0.97. Both the frame-change scores (+3.02) and topic mix (+3.09) increased the score. This issue-week had about 2,200 estimated original tweets, and Part B showed that issues with small samples receive high shift scores by chance. The week also had no shifts, so the rule had to predict one anyway. We could improve this by requiring a minimum sample size before scoring or weighting frame-change features by their noise floor.

In addition, we had a data failure that came from missing data and misleading history in our features. For example, abortion in early August was labeled a shift, but the model missed it because the issue's past weeks pulled its score down and the week was missing a noise floor. That week's tweets shifted to the Republican Party's 2024 policy platform, Trump's softer position on abortion, and IVF. The frame-change scores raised the score strongly (+4.16), but the issue's history worked against it: the lag features, which hold the shift scores from the past two weeks, lowered the score (−2.21). The week was also missing data, since it didn't have enough tweets for a noise floor, so that value was filled in with median imputation. On top of this, the row was in fold 1, which trained on only 35 rows. To improve this, we could drop the lag features since the ablation didn't show a benefit, handle missing noise floors with a better method than median imputation, and add training data.

Finally, we might have label failure, which is errors caused by noise in the labels, since we used an LLM to label our data. If a label is wrong, the model would learn from the wrong answer during training and be graded against it during testing, resulting in a correct prediction that would count as an error. In our project, our hand labels disagreed with the LLM on 1 of 10 rows, an abortion row where the shift built over two weeks, which made it hard to label as a shift or no-shift week. We could improve this by adding a "gradual shift" label or by labeling each row with two LLM prompts and a manual review and only keeping the rows where they're in agreement.

5. Discussion

Only about 13% of the tweets mentioned any of our five issues. Most of the data are candidates, reactions, and hashtags. We first wondered if this was due to our smaller 26k sample, but it held with our larger sample of 465k tweets as well. We were not able to test whether bots played a part. At 5 topics, everything was grouped by candidate, and issues only appeared once we removed names and raised the topic count. Content shift (0.40-0.63) was about three to four times the proportion shift (about 0.14), so narratives changed mostly through new wording, not attention moving between frames. I had expected the reverse. Abortion had the highest shift despite being one of the smallest issues, which may be partly noise. Attention and shift were barely related: weeks when an issue got more tweets were not necessarily the weeks it was framed differently.

Our main challenges were these: Topic models find topics, not issues, so we used keyword patterns to pull out issue tweets, fixed false matches ('nato' in 'senator,' 'roe' in 'heroes'), and hand-checked 100 tweets (precision 0.85). The full dataset (about 46 million tweets) was too large to work with directly, and our first 26k sample was too small to score weeks, so we moved to a sample of 465k.

With more time and storage, we would try BERTopic, which is less literal and could capture a tweet's context and more meaning even when no keyword appears. We would validate the shift score with human coders, since we only hand-checked the keyword tags. We would replace keyword tagging with a classifier trained on labeled tweets, since foreign policy precision was only 0.70.

5.2 Learnings from Part A

We were surprised that the raw shift score was almost the same between shift and no-shift weeks (mean 0.315 vs. 0.308), but once we subtracted the noise floor, shift weeks stood out a little better (+0.044 vs. -0.009). We also found

that shifts would cluster by weeks. For example, weeks 12 and 27 had four shifts across all five issues each, and weeks 17 and 30 didn't have any. We were also surprised that tuning made every model slightly worse. Also, the change in the best model depended on whether we averaged fold scores or pooled the test rows.

took the majority vote. We hid the dates and scores from the LLM and agreed with 9 out of 10 labels we checked by hand. We agreed on 9 of 10 of the LLM's labels. The second was having a small dataset. We only had 137 rows and 22 shifts, so one row could change a fold's score greatly. We split by week, reported standard deviations across folds, used nested cross-validation, and ran a permutation test to estimate the chance level. The third challenge was the uncalibrated scores, so we used a within-week ranking rule because a threshold tuned on the training weeks didn't transfer to the test weeks. Finally, due to our dependency on Part B, we built and tested the pipeline on placeholder data while we waited for its features. The most important thing we could do with more time is increase the rows in our dataset by reducing the time windows from weeks to days and/or including more issues. We would also add meaning-based features from sentence embeddings since our failure analysis found shifts that keep similar vocabulary. Lastly, we would calibrate scores so the number of predicted shifts could vary by week.We had four main challenges. The first was having no labels, so we had an LLM label each week three times and

6. Ethical Considerations

There are several ethical issues that are relevant to our unsupervised learning work. First, X users did not agree to be studied; however, we used a publicly released research dataset, reported only aggregated results, and did not identify or quote individual accounts. Second, X users are not representative of the population. Pew found that Twitter users skew younger, more educated, and more Democratic than U.S. adults, and that the top 10% of users produce about 80% of tweets [5], so our results describe the English-language conversation on X, not all voters. Coordinated inauthentic accounts were also active in the 2024 election conversation on X [2], and we did not filter for them. Some of the spam we saw in our hand check, such as the ROEvember follow train tweets, suggests they may have shaped some of our topics. Third, keyword tagging can mislabel tweets, so some groups' speech may be over- or under-counted. We measured precision (0.85 overall, 0.70 for foreign policy) but not recall, so we can't say how many relevant tweets we missed. Lastly, a shift detector could be misused to target people or label political speech. Our goal with this project was solely exploratory, not to create a tool for monitoring individuals. Finally, shift scores don't show who changed a narrative or why.

For supervised learning, our first ethical concern is our labels. Since they are labeled by an LLM, it's possible they may be biased in how the model judges left and right leaning framing, and our classifier would learn that bias. We could reduce this risk by manually labeling a larger sample that's balanced across issues and political stances, then report the agreement rate for each group. The second concern is that our model could be used, and it's not clearly better than chance. If it were used for policy decisions, for example, it would lead to false positives and missed shifts. For this reason, we present our results as exploratory and recommend that a human reviewer check the predictions alongside our scores. Finally, even if our model were accurate, it would only show that a narrative changed, not why. A detected shift shouldn't be used to explain why a shift happened, and this limitation should be stated wherever the results are shared.

7. Statement of Work

Jesse Kim was the lead on the data processing and also handled GitHub management and project task organization within GitHub. Tanisha-Kelly Garrett led the unsupervised analysis (LDA/NMF topic modeling, weekly shift scores, and hand check of keyword precision) and wrote Part B. Shay Gooding was the lead on the project idea, initial literature review, and the supervised analysis (LLM labeling and manual verification, volume and noise-floor features, and supervised model training and evaluation) and wrote Part A. All members assisted others as needed, contributed to and wrote sections of the final report, and reviewed the final draft.

8. References

1. Liedke, J., & St. Aubin, C. (2024, September 17). Social media and news fact sheet. Pew Research Center. https://www.pewresearch.org/journalism/fact-sheet/social-media-and-news-fact-sheet/

2. Cinus, F., Minici, M., Luceri, L., & Ferrara, E. (2025). Exposing cross-platform coordinated inauthentic activity in the run-up to the 2024 U.S. election. In Proceedings of the ACM Web Conference 2025 (pp. 541–559). Association for Computing Machinery. https://doi.org/10.1145/3696410.3714698

3. Lange, K.-R., Schmidt, T., Reccius, M., Müller, H., Roos, M., & Jentsch, C. (2025). Narrative shift detection: A hybrid approach of dynamic topic models and large language models (arXiv:2506.20269). arXiv. https://doi.org/10.48550/arXiv.2506.20269

4. Farr, D., Prochaska, S., Moody, J., Ng, L. H. X., Cruickshank, I., Starbird, K., & West, J. (2026). Temporal narrative monitoring in dynamic information environments (arXiv:2603.17617). arXiv. https://doi.org/10.48550/arXiv.2603.17617

5. Wojcik, S., & Hughes, A. (2019, April 24). Sizing up Twitter users. Pew Research Center. https://www.pewresearch.org/internet/2019/04/24/sizing-up-twitter-users/

6. Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. Advances in Neural Information Processing Systems, 30, 4765–4774. https://proceedings.neurips.cc/paper_files/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf

7. Balasubramanian, A., Zou, V., Narayana, H., You, C., Luceri, L., & Ferrara, E. (2024). A public dataset tracking social media discourse about the 2024 U.S. presidential election on Twitter/X (arXiv:2411.00376). arXiv. https://doi.org/10.48550/arXiv.2411.00376

9. Appendix

Table A1. Supervised features (17).

Group	Feature	Measures
Volume (3)	log_orig_per_day	Estimated tweets per day about the issue (log scale)
	issue_share	Issue's share of that week's sampled tweets
	issue_share_change	Change in issue_share from last week
Frame change (7)	shift_score	Average of content_shift and proportion_shift
	content_shift	How much matched frames changed their wording
	proportion_shift	How much attention moved between frames
	new_frames	Number of frames with no match last week
	shift_minus_noise	shift_score minus the noise floor
	noise_missing	1 if the week had too few tweets for a noise floor
	shift_zscore	How unusual this week's shift is for the issue
Topic mix (5)	topic_biden, topic_trump, topic_harris, topic_maga_movement, topic_partisan_talk	Share of each full-corpus NMF topic among the issue's tweets
Past weeks (2)	lag1_shift, lag2_shift	shift_score from 1 and 2 weeks earlier

Note: Target (not a feature): label, where 1 = narrative shift and 0 = no shift.